Examiner only

7. *(a)* (i) Chlorine is a non-metal found in Group 7 of the Periodic Table. When it is bubbled into a solution of potassium iodide there is a colour change from pale green to brown. Explain why this reaction occurs. [2]

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(ii) Write the balanced symbol equation for the reaction between chlorine and potassium iodide. [2]

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(b) The symbol equation for the reaction between iron and chlorine is as follows.

$$2Fe \quad + \quad 3Cl_2 \longrightarrow 2FeCl_3$$

Calculate the mass of chlorine needed to react with 1.32 g of iron. [3]

$A_r(Fe) = 56 \qquad A_r(Cl) = 35.5$

Mass of chlorine = ... g

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(c) (i) Under certain conditions, Group 7 elements will react with each other to produce new compounds.

When chlorine is reacted with bromine, chlorine tribromide is made.

Balance the symbol equation for this reaction. [1]

$$Cl_2 \quad + \quad \boxed{}\, Br_2 \longrightarrow \boxed{}\, ClBr_3$$

(ii) A chemist calculated that if she reacted 7.00 g of chlorine with an excess of bromine, the theoretical mass of chlorine tribromide produced is 27.55 g.

However, when she carried out the experiment using 7.00 g of chlorine the mass of chlorine tribromide obtained was 21.34 g.

Calculate the percentage yield of chlorine tribromide. [1]

Percentage yield = ... %

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